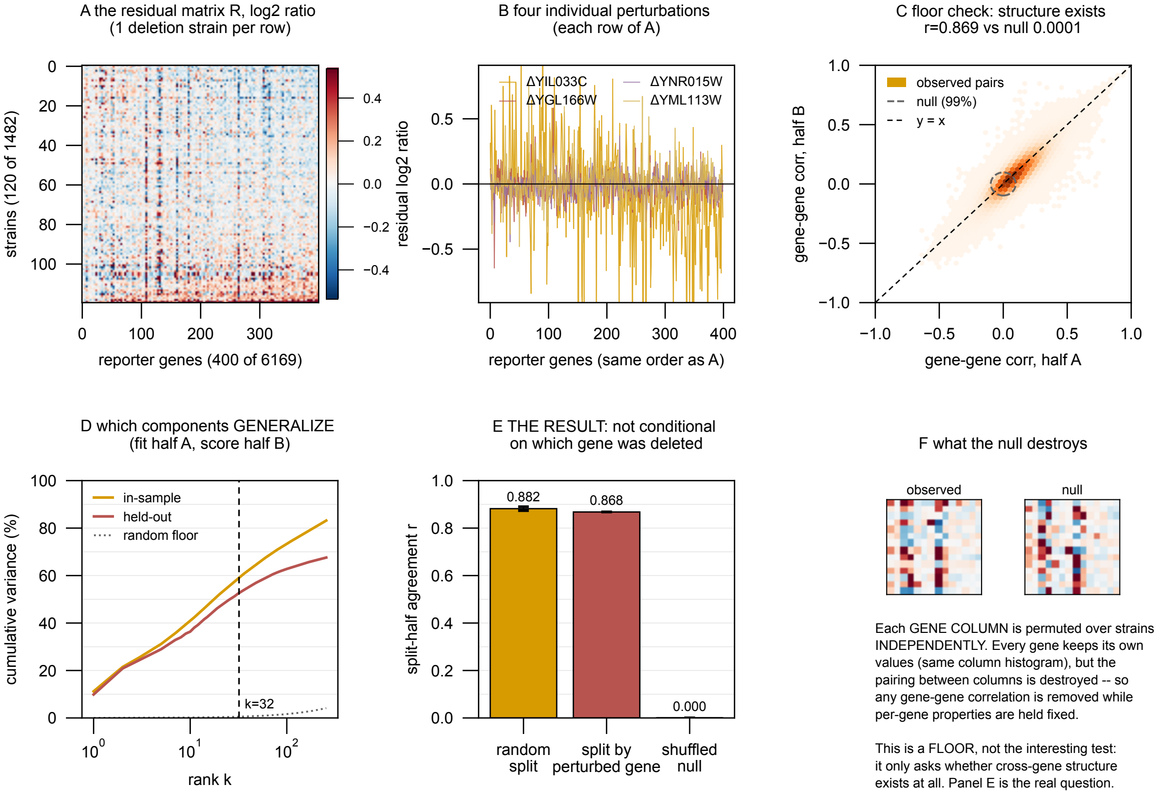
A the residual matrix R, log2 ratio
(1 deletion strain per row)
strains (120 of 1482)
reporter genes (400 of 6169)
residual log2 ratio
B four individual perturbations
(each row of A)
ΔYIL033C
ΔYGL166W
ΔYNR015W
ΔYML113W
reporter genes (same order as A)
C floor check: structure exists
r=0.869 vs null 0.0001
observed pairs
null (99%)
y = x
gene-gene corr, half B
gene-gene corr, half A
D which components GENERALIZE
(fit half A, score half B)
in-sample
held-out
random floor
k=32
cumulative variance (%)
rank k
E THE RESULT: not conditional
on which gene was deleted
0.882
0.868
0.000
split-half agreement r
random split
split by perturbed gene
shuffled null
F what the null destroys
observed
null
Each GENE COLUMN is permuted over strains INDEPENDENTLY. Every gene keeps its own values (same column histogram), but the pairing between columns is destroyed -- so any gene-gene correlation is removed while per-gene properties are held fixed.
This is a FLOOR, not the interesting test: it only asks whether cross-gene structure exists at all. Panel E is the real question.